

Isolating the cause of quantum-kernel failure on malware family attribution

Complete classical baselines, subsample validation,
kernel diagnostics and paired significance tests

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Abstract

Background. A fidelity-kernel quantum support vector machine (QSVM) had been observed to fall to the chance level on the 15-class task of CIC-MalMem-2022 while performing well above chance on the equivalent EMBER 2018 task. Class imbalance and qubit count had been excluded, and the cause was unidentified. The classical arm of the comparison on EMBER used a single model, and no difference between the two frameworks had been tested statistically.

Methods. We complete the EMBER classical baseline with four models at $n_{\text{components}} \in \{1, 3, 6\}$ on both tasks, validate the 1000-row subsample on which all quantum results rest, measure class separability of the shared PCA projection, and measure kernel concentration and kernel-target alignment against a classical RBF control built on the identical feature matrix. Every quantum–classical pair with a complete cross-validation sweep is tested with a paired t -test.

Results. Random forest beats SVM on every EMBER binary configuration, so the previously reported classical advantage was understated by 0.011 to 0.027 macro-F1. Both subsamples are faithful, with Kolmogorov–Smirnov rejection counts far below their null expectations. EMBER’s families are $\approx 8\times$ more separable than CIC’s, which accounts for poor absolute performance but not for a QSVM-specific deficit. The RBF control produces the same off-diagonal Gram spread on both corpora (0.2212 against 0.2184), whereas the fidelity kernel matches RBF on EMBER ($0.92\times$) and collapses to $0.40\times$ on CIC. All 84 comparable pairs favour the classical model at $p < 0.05$.

Conclusions. The QSVM’s collapse on CIC is attributable to kernel concentration rather than to feature difficulty, and it is a property of the kernel rather than of the corpus.

1 Introduction

A fidelity-kernel QSVM had been compared against classical baselines on two malware corpora under a protocol that equalises rows, cross-validation folds and input dimensionality between the two arms. Three questions were left open by that comparison.

First, the EMBER classical arm used a support vector machine only, so every recorded margin there was a comparison against one baseline rather than against the strongest available. Second, the QSVM fell to the chance level on CIC 15-class while reaching 0.42–0.52 macro-F1 on EMBER 15-class, and the cause was unidentified after class imbalance and qubit count had been excluded. Third, every claim of a classical advantage rested on differences in mean macro-F1 with no test that any gap exceeded fold-to-fold noise. A fourth, quieter issue was that all quantum results are computed on a 1000-row subsample whose fidelity to its population had been assumed rather than measured.

This report addresses all four. Its central contribution is a diagnosis: by building a classical RBF kernel on the identical feature matrix, we separate the hypothesis that CIC’s features are hard from the hypothesis that the fidelity kernel fails on them, and find for the latter.

2 Materials and methods

2.1 Data

Two corpora, each with a binary detection task and a 15-class family attribution task.

CIC-MalMem-2022

Obfuscated-malware memory dumps, 55 numeric forensic features. Binary: all 58 596 rows, balanced. 15-class: 29 298 malware-only rows over 15 families, $1.7\times$ imbalanced.

EMBER 2018

Static portable-executable features, 2381 dimensions, on an 18 014-row extract of the published test set. Binary: all 18 014 rows, 88.9% malware. 15-class: 14 325 rows balanced at 955 per family.

2.2 Shared protocol

Both frameworks consume one feature pipeline: a zero-variance filter, a correlation filter at 0.95, standardisation, then PCA to $n_{\text{components}}$ components, fit on training-fold rows only. $n_{\text{components}}$ is the single alignment point, setting the classical feature count and the quantum qubit budget simultaneously.

The fidelity kernel costs one circuit evaluation per pair of training samples, so quantum runs use a stratified 1000-row subsample. That subsample and its five stratified outer folds are written to disk by the quantum runner and reloaded by the classical runner, so both are scored on identical rows and folds, with 800 training and 200 held-out rows per fold.

All results are held-out macro-F1 averaged over the five outer folds. Hyperparameters are tuned by an Optuna inner loop inside each training fold. Accuracy is not reported: on EMBER binary at $n_{\text{components}} = 1$ the QSVM attains 0.890 accuracy in three of five folds while the benign class’s F1 is exactly 0.000, because 88.9% of rows are malware and the model labels everything malware.

2.3 Measurements introduced here

Complete classical baseline

Random forest, XGBoost and LightGBM added alongside the existing SVM at $n_{\text{components}} \in \{1, 3, 6\}$ on both EMBER tasks, replaying the persisted quantum splits.

Subsample representativeness

Per-class proportion drift, per-feature mean and standard-deviation drift, and a per-feature two-sample Kolmogorov–Smirnov test of the 1000-row subsample against its full population, with the rejection count read against its null expectation $\alpha \cdot n_{\text{features}}$.

Class separability

One-vs-rest Fisher discriminant ratio, silhouette coefficient, pairwise centroid distance and cumulative explained variance of the shared PCA projection, fit on training-fold rows only.

Kernel diagnostics

Off-diagonal standard deviation of the training Gram (concentration) and Frobenius kernel-target alignment $\langle K, YY^T \rangle_F / (\|K\|_F \|YY^T\|_F)$, measured for the fidelity kernel and for an RBF kernel built on the identical post-PCA feature matrix.

Paired significance

A paired t -test and a Wilcoxon signed-rank test over the five per-fold macro-F1 values for every quantum–classical pair with a complete sweep.

2.4 Selection of comparable runs

Runs are grouped by cross-validation sweep rather than by configuration. The same configuration was swept more than once, so a parameter-based grouping returns 10, 15 or 20 records for a five-fold sweep and averages distinct hyperparameter searches together; 57 parameter groups in the experiment log have a size other than five. For the QSVM the encoding is additionally a *per-fold outcome* rather than a configuration key, because the tuner selects it inside each fold, so one sweep tuning over both encodings can appear as four angle folds and one IQP fold. Fold records are therefore grouped by parent run, retaining only sweeps that completed with exactly five folds. Under that rule the log holds 125 nested sweeps, 123 of them complete.

3 Results

3.1 The classical baseline was incomplete

Table 1: EMBER 2018, held-out macro-F1, mean over five outer folds. Best classical model in bold.

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
<i>Binary</i>						
$n_{\text{components}} = 1$	0.5582	0.5451	0.5093	0.5314	0.4532	0.4587
$n_{\text{components}} = 3$	0.6575	0.6465	0.6434	0.6306	0.4924	0.5084
$n_{\text{components}} = 6$	0.6835	0.6680	0.6593	0.6724	0.5138	0.5137
<i>15-class</i>						
$n_{\text{components}} = 1$	0.5562	0.5313	0.5410	0.5294	0.1688	0.1541
$n_{\text{components}} = 3$	0.7194	0.7089	0.7035	0.7389	0.3908	0.4797
$n_{\text{components}} = 6$	0.7903	0.7659	0.7709	0.7930	0.4232	0.5195

Random forest beats SVM on every binary configuration (Table 1), so the classical advantage widens from +0.0727 to +0.0995 at $n_{\text{components}} = 1$, from +0.1222 to +0.1491 at $n_{\text{components}} = 3$, and from +0.1587 to +0.1697 at $n_{\text{components}} = 6$. On the 15-class task SVM is the strongest baseline at $n_{\text{components}} \in \{3, 6\}$, so those margins are unchanged. Four of six configurations move, and all four move in the same direction.

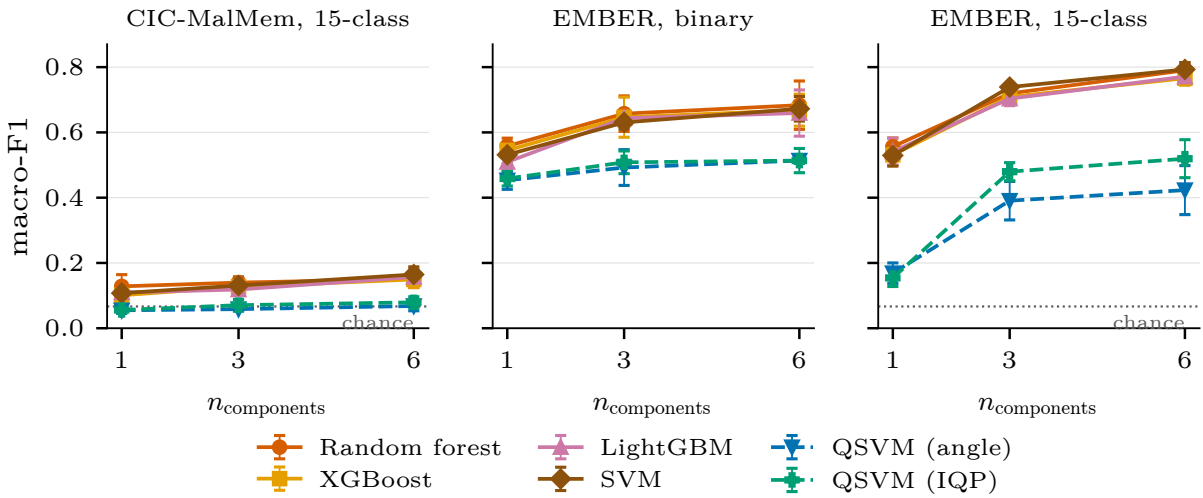


Figure 1: Macro-F1 against the shared feature and qubit budget. Solid lines are classical, dashed lines the two QSVM encodings. Error bars are one standard deviation over the five outer folds, and the dotted line marks the 15-class chance level. CIC binary is not shown because its sweeps are not separable (§4.2).

Figure 1 shows the trend with capacity. On EMBER binary the QSVM gains 0.055 macro-F1 from one to six components where the random forest gains 0.125, so the gap widens rather than closing.

3.2 The subsample preserves its population

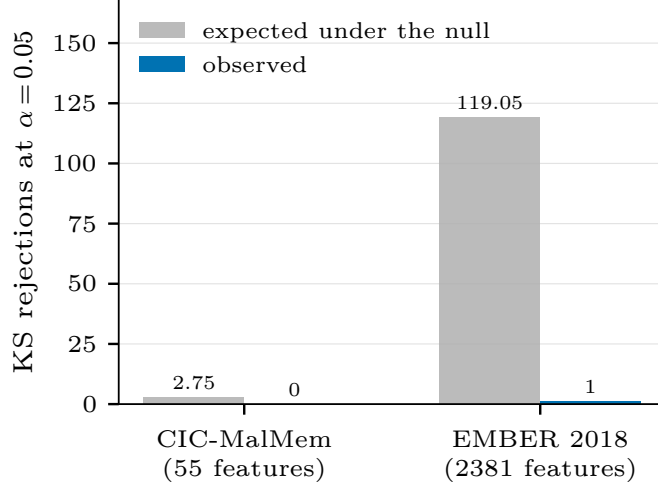


Figure 2: Per-feature Kolmogorov–Smirnov rejections at $\alpha = 0.05$, 1000-row subsample against full population, beside the count chance alone produces.

A rejection count is interpretable only against its null expectation, since a faithful subsample still rejects about $\alpha \cdot n_{\text{features}}$ features by chance. CIC rejects 0 of 55 features where ≈ 2.75 were expected, and EMBER 1 of 2381 where ≈ 119 were expected (Figure 2). Maximum class-proportion drift is below 0.001 on both, and maximum per-feature mean drift is 0.046 and 0.122 population standard deviations respectively.

3.3 Feature geometry accounts for the corpus, not the kernel

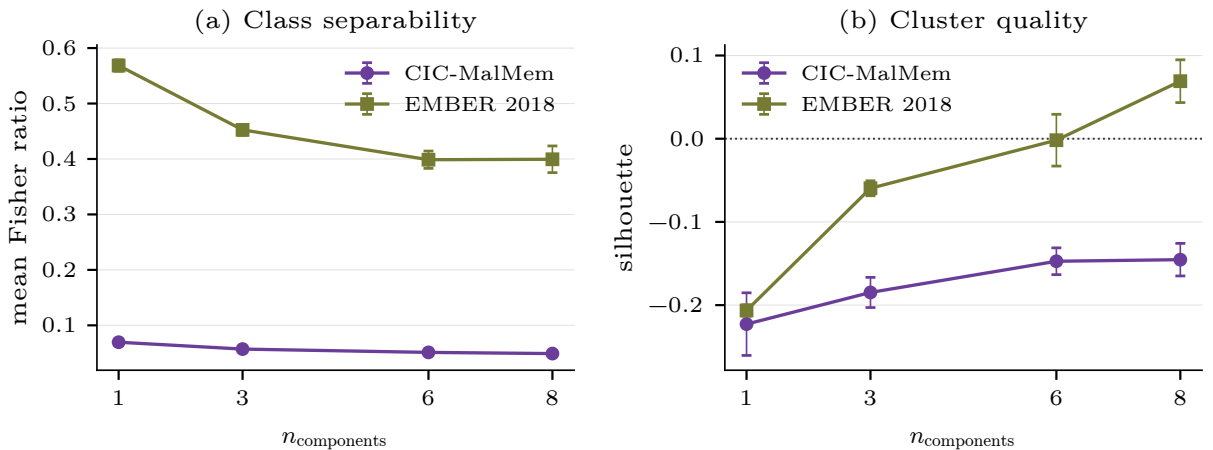


Figure 3: Class separability of the shared PCA projection, averaged over the five outer folds. Both metrics are scale-invariant and therefore comparable across corpora of very different dimensionality.

EMBER’s families are $\approx 8\times$ more separable by one-vs-rest Fisher ratio and up to $50\times$ on the worst-case family (Figure 3). CIC’s silhouette coefficient is negative at every $n_{\text{components}}$, so the average point lies closer to another family’s centroid than to its own. CIC’s PCA nonetheless

explains substantially more variance (33–81% against 8–30%), placing its variance in directions that do not discriminate family.

This accounts for the poor absolute performance of every method on CIC 15-class. It does not account for a QSVM-specific deficit, since a classical SVM extracts 0.108–0.165 macro-F1 from these same features where the QSVM extracts 0.056–0.079.

3.4 Kernel concentration accounts for the kernel

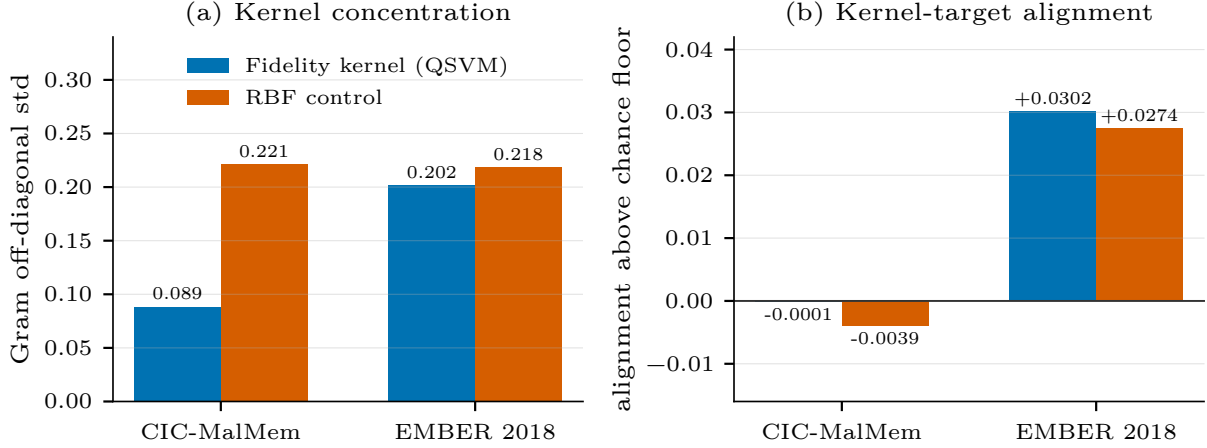


Figure 4: First outer training fold, $n_{\text{components}} = 6$, IQP encoding, 800 rows, with an RBF control built on the identical feature matrix. (a) Off-diagonal Gram spread. (b) Kernel-target alignment relative to its floor: a constant kernel scores $1/\sqrt{15} \approx 0.258$ on 15 balanced classes, so only the excess is informative.

The RBF control produces near-identical off-diagonal spread on both corpora, 0.2212 against 0.2184, so CIC’s features do not cause a classical kernel to concentrate. The fidelity kernel tracks RBF on EMBER at $0.92\times$ and collapses to $0.40\times$ on CIC, compressing every off-diagonal similarity into a band $2.5\times$ narrower from the same inputs and driving the Gram toward a scaled identity.

Alignment behaves differently. On CIC the fidelity kernel lands 0.000069 *below* the constant-kernel floor, making it indistinguishable from a matrix of ones, and the RBF control is also below it; on EMBER both clear the floor by ≈ 0.03 . Alignment therefore separates the two corpora while concentration separates the two kernels.

Table 2: Off-diagonal Gram spread across the 15-class sweep, mean over five outer folds. These values were recorded on every fit, so no additional computation was required.

	$n_{\text{components}} = 1$	$n_{\text{components}} = 3$	$n_{\text{components}} = 6$	$n_{\text{components}} = 8$
CIC, angle	0.1192	0.0795	0.0674	0.0600
CIC, IQP	0.0927	0.1005	0.0882	0.0798
EMBER, angle	0.2044	0.1942	0.1363	—
EMBER, IQP	0.1756	0.2569	0.2104	—

Table 2 confirms the effect is not an artefact of one fold or one setting: EMBER’s Gram is $1.9\text{--}2.6\times$ more spread than CIC’s at every comparable cell, and concentration worsens with qubit count on both, which is the trend the bandwidth parameter exists to counteract.

3.5 The differences are statistically significant

All 84 comparable pairs favour the classical model and all 84 reach $p < 0.05$. The largest p -value across every pair is 0.0241 and the smallest is 4×10^{-6} .

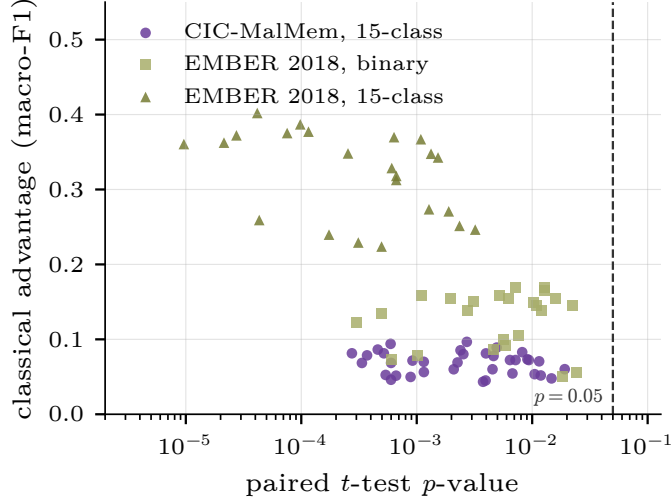


Figure 5: Effect size against paired t -test p -value for all 84 comparable quantum-classical pairs. Every point lies left of $p = 0.05$ and above zero.

A Wilcoxon signed-rank test returns exactly 0.0625 for all 84 pairs. With five paired samples that is the smallest attainable two-sided p -value, so the test cannot reach 0.05 at this fold count regardless of effect size; all 84 reaching the floor indicates that every pair achieved the most extreme rank configuration available.

Among classical models on shared held-out predictions, McNemar’s test separates random forest from XGBoost on EMBER 15-class at $p = 0.0022$, while on CIC 15-class no pair of classical models separates at all, consistent with the geometry result above.

4 Discussion

4.1 Interpretation

Three of the four open questions resolve cleanly. The classical baseline was incomplete in a direction that understated the classical advantage; the subsample underpinning every quantum result is faithful to its population; and the observed differences are significant on every comparable pair rather than resting on mean differences alone.

The fourth is the substantive finding. Feature geometry and kernel geometry give different answers, and the difference is the diagnosis. Geometry explains why every method performs poorly on CIC 15-class, since the projection places class centroids closer to foreign points than to their own. It cannot explain a deficit specific to the QSVM, because the classical SVM recovers two to three times the QSVM’s macro-F1 from the same projection. Kernel concentration does explain it: the fidelity kernel loses 60% of the RBF control’s off-diagonal spread on inputs where the control loses none. A Gram approaching a scaled identity presents the support vector machine with a geometry in which all pairs are equally similar, which is what a macro-F1 pinned to chance looks like internally.

The result is a statement about the method rather than the corpus, and it predicts that the approach will degrade wherever the encoded states become mutually near-orthogonal.

4.2 Limitations

Bandwidth is untested on CIC

Concentration is the identified mechanism and bandwidth is the parameter that acts on it, but no bandwidth sweep was run. §3.4 identifies a mechanism; it does not demonstrate a remedy.

Alignment is a weak predictor

Alignment reports that neither kernel carries label information on CIC, yet a classical SVM reaches 0.16 there. Alignment is a global, unweighted average over all pairs, and a support vector machine requires no such global property. It ranks corpus difficulty usefully and predicts attainable SVM performance poorly. The concentration measurement does not share this weakness, since it describes the matrix the SVM actually receives.

Scope of the kernel measurement

The alignment and RBF-control figures come from one training fold, one capacity and one encoding. Table 2 is broader, spanning five folds, four capacities and both encodings, but carries no RBF control. The RBF control itself uses a single unturned γ heuristic, so the comparison is against a reasonable classical default rather than the best attainable classical kernel.

One configuration is not separable

Every CIC binary run, classical and quantum alike, predates sweep-level tracking in the experiment log and carries no recoverable sweep boundary; 45 fold records per model resolve into two distinct sweeps per $n_{\text{components}}$ with no identifier separating them. No grouping was reconstructed, since the time gaps between runs are not a recorded boundary. The limitation is symmetric across frameworks, and it is why CIC binary is absent from Figure 1 and from the 84 tested pairs.

No sample-level test between frameworks

McNemar’s test requires per-sample predictions from both models on a shared test set. QSVM predictions were not persisted during the reported sweeps, so cross-framework evidence rests on five paired fold-level scores.

Fold count

Five outer folds support the paired t -test and are too few for Wilcoxon, whose minimum attainable p -value at $n = 5$ is 0.0625. Six folds would place 0.03125 within reach.

Simulator only

All quantum results use a state-vector simulator, with no hardware execution and no noise model.

4.3 Future work

Sweep bandwidth on CIC, since the mechanism is identified and the corresponding parameter has never been tuned there beyond its default. Move to six outer folds so that a non-parametric test becomes usable.

5 Conclusion

Completing the classical baseline widened the recorded classical advantage on EMBER binary by 0.011 to 0.027 macro-F1, and paired testing establishes that all 84 comparable quantum–classical differences reach $p < 0.05$. The 1000-row subsample on which every quantum result depends is faithful to its population.

The QSVM’s collapse to chance on CIC 15-class is attributable to kernel concentration rather than to feature difficulty. A classical RBF kernel built on the identical feature matrix retains its off-diagonal spread while the fidelity kernel falls to $0.40\times$ of it, which locates the failure in the kernel and names the parameter that acts on it.